

## No. of Functional Features Included in the Solution

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          | 5 July 2026                                       |
| <b>Team ID</b>       | SWTID-2026-4860                                   |
| <b>Project Name</b>  | EduGenie Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                           |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                          | Feature Description                                                                             | Module / Component                                      | Status (Done / In Progress / Pending) | Marks Contribution |
|------|---------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------|--------------------|
| 1    | Intelligent Question Answering        | Provides detailed, context-aware answers to student and educator questions.                     | Backend Endpoint: /qa (Cloud API Wrapper)               | Done                                  |                    |
| 2    | Simplified Concept Explanations       | Generates simplified, age-appropriate explanations using custom tailored educational analogies. | Backend Endpoint: /explain (Cloud API Wrapper)          | Done                                  |                    |
| 3    | AI-Powered Quiz Generation            | Creates custom multiple-choice test questions on specific academic topics with answers.         | Backend Endpoint: /quiz (Cloud API Wrapper)             | Done                                  |                    |
| 4    | Educational Text Summarization        | Performs fast, local text summaries to extract core points from user notes or text passages.    | Backend Endpoint: /summarize (Local Inference Pipeline) | Done                                  |                    |
| 5    | Personalized Learning Recommendations | Builds comprehensive, step-by-step sequential learning roadmaps tailored to study goals.        | Backend Endpoint: /recommend (Cloud API Wrapper)        | Done                                  |                    |
| 6    | Single-Page Tab Workspace             | Interactive dashboard allowing users to switch tabs smoothly across all core tools.             | Frontend Layout: templates/index.html                   | Done                                  |                    |

### Feature Summary:

| Metric                     | Count / Value                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Features Planned     | 6                                                                                                                                                          |
| Total Features Implemented | 6                                                                                                                                                          |
| Core / Must-Have Features  | <ul style="list-style-type: none"> <li>Intelligent Q&amp;A</li> <li>Text Summarizer</li> <li>Concept Simplifier</li> <li>User Form UI Dashboard</li> </ul> |

|                                    |                                                                                                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Additional / Nice-to-Have Features | <ul style="list-style-type: none"><li>• AI-Powered Quiz Generation Engine</li><li>• Personalized Path Recommendation System</li></ul>                                                                  |
| Features Tested & Verified         | <ul style="list-style-type: none"><li>• All 5 AI functional processing endpoints validated without errors.</li><li>• Secure terminal environment variables configuration structure verified.</li></ul> |

Feature Category Breakdown:

| S.No | Category                  | Features in Category                                                                                                                                                         | Example Features                                                                                                                                                                                |
|------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Interface (UI)       | <ul style="list-style-type: none"><li>• Single-page responsive tabs workspace interface</li><li>• Dynamic answer/output display sections</li></ul>                           | <ul style="list-style-type: none"><li>• templates/index.html</li><li>• static/style.css</li><li>• CSS @keyframes fadeIn animations</li></ul>                                                    |
| 2    | Backend / Logic           | <ul style="list-style-type: none"><li>• Asynchronous request routing execution system</li><li>• Target prompt engineering formulation templates</li></ul>                    | <ul style="list-style-type: none"><li>• FastAPI router app object instance</li><li>• handle_explain tailored analogies prompt</li><li>• Hugging Face pipeline("text2text-generation")</li></ul> |
| 3    | Database / Storage        | <ul style="list-style-type: none"><li>• Ephemeral context dictionary passing engines</li><li>• Static static file storage path directories</li></ul>                         | <ul style="list-style-type: none"><li>• Jinja2 dynamic template rendering context</li><li>• app.mount("/static", ...) configuration</li></ul>                                                   |
| 4    | API / Integration         | <ul style="list-style-type: none"><li>• Cloud artificial intelligence SDK orchestration layers</li><li>• Hardware optimized local inference pipeline model weights</li></ul> | <ul style="list-style-type: none"><li>• from google import genai model connection</li><li>• MBZUAI/LaMini-Flan-T5-248M pipeline</li></ul>                                                       |
| 5    | Security / Authentication | <ul style="list-style-type: none"><li>• Secure configuration system hidden from code files</li><li>• Pre-runtime validation logic constraints and handlers</li></ul>         | <ul style="list-style-type: none"><li>• os.getenv("GEMINI_API_KEY") parameter wrapper</li><li>• Missing credential key validation ValueError crash check</li></ul>                              |